



**Stan's
Technologies**

Benchmark Report

BoolMinGeo vs PyEDA Boolean Minimization

Experiment Date: 2026-01-15

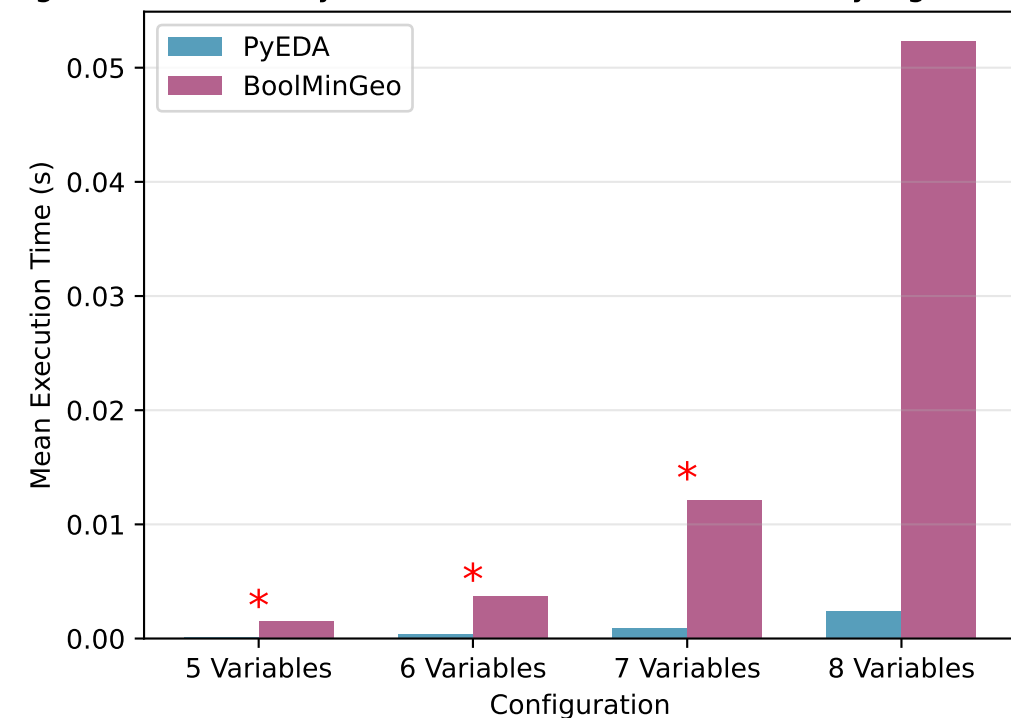
Random Seed: 42

Total Test Cases: 40

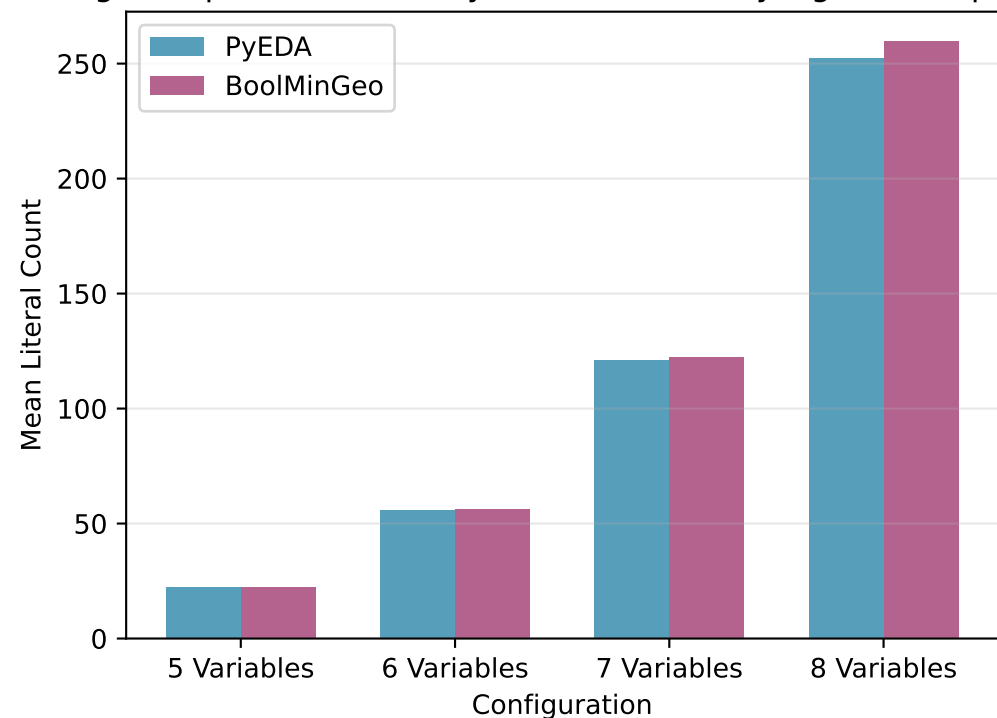
Statistical Significance Level: $\alpha = 0.05$

A Statistical Comparison of Boolean Minimization Algorithms

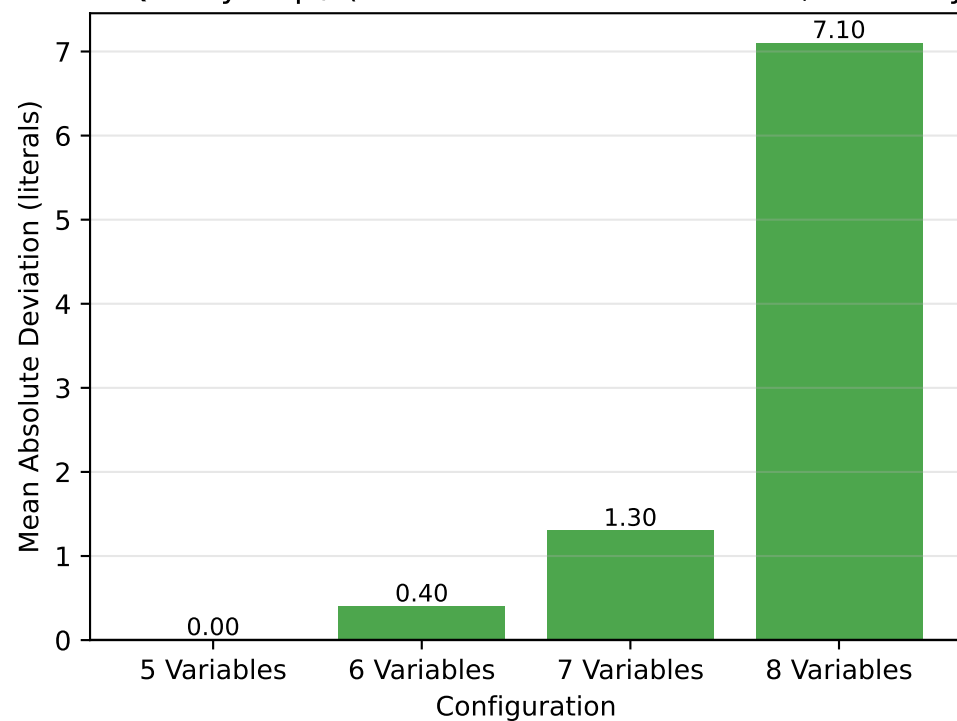
Average Performance by Variable Count\n(* = statistically significant, p < 0.05)



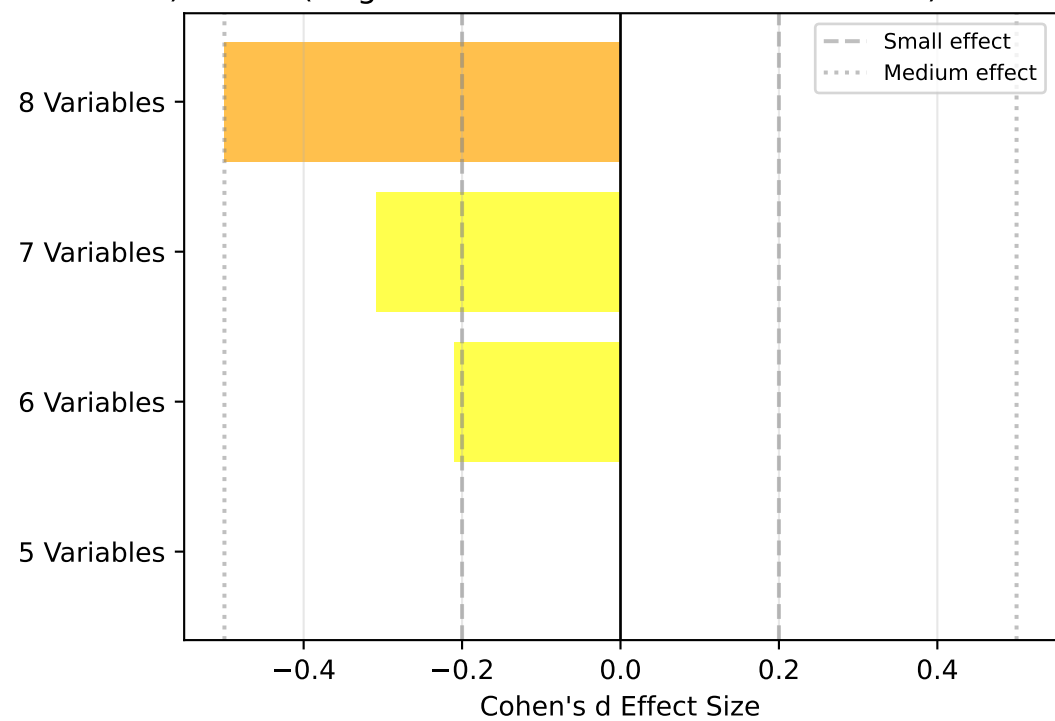
Average Simplification Quality\n(* = statistically significant, p < 0.05)



Simplification Quality Gap\n(Green = BoolMinGeo better, Red = PyEDA better)



Effect Size: Simplification Quality
(Negative = BoolMinGeo more minimal)



STATISTICAL ANALYSIS SUMMARY

EXPERIMENT OVERVIEW

Total Test Cases:	40
Configurations Tested:	4 (5-8 variables)
Tests per Config:	10 (sparse, balanced, dense distributions)
Random Seed:	42
Significance Level:	$\alpha = 0.05$

5-VARIABLE FUNCTIONS (10 tests, 10 non-constant)

EXECUTION TIME:

Mean PyEDA:	0.000164 s
Mean BoolMinGeo:	0.001516 s
Mean Difference:	-0.001352 s
95% CI:	[-0.001691, -0.001013]
p-value:	0.000008
Statistically Sig:	YES
Effect Size (d):	-2.8523 (large)
Winner:	PyEDA

SIMPLIFICATION QUALITY (Literal Count):

Mean PyEDA:	22.50 literals
Mean BoolMinGeo:	22.50 literals
Mean Difference:	+0.00 literals
Mean Deviation:	0.00 literals
Quality Gap:	0.00% relative to PyEDA
95% CI:	[nan, nan]
p-value:	nan
Statistically Sig:	NO
Effect Size (d):	0.0000 (negligible)
Winner:	TIE

6-VARIABLE FUNCTIONS (10 tests, 10 non-constant)

EXECUTION TIME:

Mean PyEDA:	0.000351 s
Mean BoolMinGeo:	0.003710 s
Mean Difference:	-0.003358 s
95% CI:	[-0.004579, -0.002137]
p-value:	0.000155
Statistically Sig:	YES
Effect Size (d):	-1.9676 (large)
Winner:	PyEDA

SIMPLIFICATION QUALITY (Literal Count):

Mean PyEDA:	55.70 literals
Mean BoolMinGeo:	56.10 literals
Mean Difference:	-0.40 literals
Mean Deviation:	0.40 literals
Quality Gap:	0.72% relative to PyEDA
95% CI:	[-1.76, 0.96]
p-value:	0.521707
Statistically Sig:	NO
Effect Size (d):	-0.2108 (small)
Winner:	PyEDA

7-VARIABLE FUNCTIONS (10 tests, 10 non-constant)

EXECUTION TIME:

Mean PyEDA:	0.000888 s
Mean BoolMinGeo:	0.012164 s
Mean Difference:	-0.011276 s
95% CI:	[-0.016239, -0.006313]
p-value:	0.000612
Statistically Sig:	YES
Effect Size (d):	-1.6252 (large)
Winner:	PyEDA

SIMPLIFICATION QUALITY (Literal Count):

Mean PyEDA:	121.10 literals
Mean BoolMinGeo:	122.40 literals
Mean Difference:	-1.30 literals
Mean Deviation:	1.30 literals
Quality Gap:	1.07% relative to PyEDA
95% CI:	[-4.32, 1.72]
p-value:	0.355179
Statistically Sig:	NO
Effect Size (d):	-0.3082 (small)
Winner:	PyEDA

8-VARIABLE FUNCTIONS (10 tests, 10 non-constant)

EXECUTION TIME:

Mean PyEDA:	0.002378 s
Mean BoolMinGeo:	0.052293 s
Mean Difference:	-0.049915 s
95% CI:	[-0.077966, -0.021864]
p-value:	0.002994
Statistically Sig:	YES
Effect Size (d):	-1.2729 (large)
Winner:	PyEDA

SIMPLIFICATION QUALITY (Literal Count):

Mean PyEDA:	252.50 literals
Mean BoolMinGeo:	259.60 literals
Mean Difference:	-7.10 literals
Mean Deviation:	7.10 literals
Quality Gap:	2.81% relative to PyEDA
95% CI:	[-17.24, 3.04]
p-value:	0.147723
Statistically Sig:	NO
Effect Size (d):	-0.5008 (medium)
Winner:	PyEDA

CONCLUSIONS

KEY FINDINGS

- 1. PERFORMANCE:
 - Average PyEDA time: 0.000945 s
 - Average BoolMinGeo time: 0.017421 s
 - Difference: -0.016475 s
 - Statistical significance: YES (p = 0.000538)
- 2. SIMPLIFICATION QUALITY:
 - Average PyEDA literals: 112.95
 - Average BoolMinGeo literals: 115.15
 - Mean deviation: 2.20 literals
 - Quality gap: 1.95% relative to PyEDA
 - Statistical significance: NO (p = 0.079602)
- 3. EQUIVALENCE:
 - All 40/40 tests passed logical equivalence checks
 - Both algorithms produce functionally correct results
- 4. EFFECT SIZES:
 - Performance: -0.5964 (medium)
 - Simplification: -0.2846 (small)

INTERPRETATION

Per-Variable Analysis:

- 5 variables: 0.00% quality gap, Neither produces more minimal results (Mean deviation: 0.00 literals)
- 6 variables: 0.72% quality gap, PyEDA produces more minimal results (Mean deviation: 0.40 literals)
- 7 variables: 1.07% quality gap, PyEDA produces more minimal results (Mean deviation: 1.30 literals)
- 8 variables: 2.81% quality gap, PyEDA produces more minimal results (Mean deviation: 7.10 literals)

REPRODUCIBILITY

- Random seed: 42
- All test cases can be reproduced with this seed
- Statistical tests use standard methods (paired t-test, Cohen's d)

Report generated: 2026-01-15 15:08:15

© Stan's Technologies 2026